

Vector Calculus

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§1 Preliminaries

§1.1 Vector notation

We will in general work in $\mathbb{R}^3$, but sometimes $\mathbb{R}^2$. In fact, most of our results will generalise to $\mathbb{R}^n$.

We will write column vectors to save space when we mean row vectors, and we will use dot notation for differentiation (even though t isn't really time).

If $\mathbf{a}$ and $\mathbf{b}$ are vectors, then the following hold:

$$\dot{\mathbf{a}}(t) = \frac{d\mathbf{a}}{dt} = \frac{d}{dt}(a_1, a_2, a_3) = (\dot{a}_1, \dot{a}_2, \dot{a}_3) \quad (1)$$

$$\frac{d}{dt}(\mathbf{a} \cdot \mathbf{b}) = \dot{\mathbf{a}} \cdot \mathbf{b} + \mathbf{a} \cdot \dot{\mathbf{b}} \quad (2)$$

$$\frac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \dot{\mathbf{a}} \times \mathbf{b} + \mathbf{a} \times \dot{\mathbf{b}} \quad (3)$$

and these results can be shown using suffix notation. Integration is also componentwise. We have therefore

$$\int_{t_1}^{t_2} \dot{\mathbf{a}}(t) dt = \mathbf{a}(t_2) - \mathbf{a}(t_1)$$

The modulus of a vector will be denoted by the same letter as the vector, but not bolded/underlined. So, for example, $r = |\mathbf{r}|$ (but x means x_1 , not $|\mathbf{x}|$).

Notationally, we might write ∂_i for $\frac{\partial}{\partial x_i}$.

§1.2 The chain rule

Recall that, if f is a function of $\mathbf{x}$ which is itself a function of t , then

$$\begin{aligned}\frac{d}{dt}f(\mathbf{x}(t)) &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \\ &= \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}\end{aligned}$$

If instead $\mathbf{x} = \mathbf{x}(\mathbf{u})$, where $\mathbf{u} = (u_1, u_2, u_3) = (u, v, w)$, then

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u}$$

and similarly for $\frac{\partial f}{\partial v}$ and $\frac{\partial f}{\partial w}$, i.e.

$$\frac{\partial f}{\partial u_i} = \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial u_i}$$

§1.3 Taylor's theorem

Recall that for a small vector displacement $\delta\mathbf{x}$,

$$\begin{aligned}f(\mathbf{x} + \delta\mathbf{x}) &= f(x + \delta x, y + \delta y, z + \delta z) \\ &= f(x, y, z) + \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial z}\delta z + \frac{1}{2!} \left(\delta x \frac{\partial}{\partial x} + \delta y \frac{\partial}{\partial y} + \delta z \frac{\partial}{\partial z} \right)^2 f + \dots\end{aligned}$$

or, in suffix notation,

$$f(\mathbf{x} + \delta\mathbf{x}) = f(\mathbf{x}) + \frac{\partial f}{\partial x_i} \delta x_i + O(|\delta\mathbf{x}|^2)$$

§1.4 Infinitesimals

In one dimension, we have from Taylor's theorem

$$\delta f = f(x + \delta x) - f(x) = f'(x)\delta x + O(\delta x^2)$$

Taking $\delta x \rightarrow 0$, δx and δf become dx and df , and we get the exact equality

$$df = f'(x) dx$$

If we want, we can even think of infinitesimals as numbers such that their squares are zero.

Note that infinitesimals have their own product and quotient rules, namely

$$\begin{aligned}d(fg) &= g df + f dg \\ d\left(\frac{f}{g}\right) &= \frac{g df - f dg}{g^2}\end{aligned}$$

We also get

$$df = \frac{\partial f}{\partial x_i} dx_i$$

The modulus of an infinitesimal vector can be calculated using, for example

$$|d\mathbf{x}| = |dx| \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2}$$

If dx and dy are independent infinitesimals then $dx dy$ is an infinitesimal area. But if y depends on x in some way then $dx dy$ vanishes, since it's proportional to dx^2 .

§2 Vector differential operators

Remark. What is an operator? It is a map between vector spaces of functions.

§2.1 Del

The fundamental differential operator is *del*:

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

We can also write

$$\nabla = \mathbf{e}_1 \frac{\partial}{\partial x_1} + \mathbf{e}_2 \frac{\partial}{\partial x_2} + \mathbf{e}_3 \frac{\partial}{\partial x_3} = \mathbf{e}_i \frac{\partial}{\partial x_i}$$

where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is the canonical basis for $\mathbb{R}^3$.

§2.2 Gradient

When del is applied to a scalar field $f(\mathbf{x})$, it is known as the *gradient* operator:

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

This can also be denoted $\text{grad} f$.

In suffix notation, we have

$$\begin{aligned} (\nabla f)_i &= \frac{\partial f}{\partial x_i} \\ \iff \nabla f &= \frac{\partial f}{\partial x_i} \mathbf{e}_i \end{aligned}$$

Example 2.1 • If $f(\mathbf{x}) = |\mathbf{x}|^2 = x^2 + y^2 + z^2$, then $\nabla f = (2x, 2y, 2z) = 2\mathbf{x}$.
Alternatively, $\frac{\partial}{\partial x_i}(x_j x_j) = 2x_i \delta_{ij} = 2x_i$ (the δ_{ij} comes from $\frac{\partial x_j}{\partial x_i}$).